

Unit 5. Covariance, Conditional Expectation and Random Vectors

Covariance, correlation, mean vector & covariance matrix, CEF, LIE

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 13–14, CC BY-NC-SA 4.0.

Roadmap

- 1 Covariance
- 2 Correlation coefficient
- 3 Random vectors
- 4 Conditional expectation
- 5 Law of iterated expectations
- 6 Population OLS
- 7 Wrap-up

Covariance: how do X and Y move together?

Definition

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Properties

- $\text{Cov}(X, X) = \text{Var}(X)$.
- Symmetric: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
- **Computational**: $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.
- Bilinear in each argument.
- $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$.
- $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$. Converse is **false**.

Variance of a sum

Theorem

For any X, Y :

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

General formula:

$$\text{Var}\left(\sum_i a_i X_i\right) = \sum_i a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

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Risk implication. Two assets with the same variance combine to a portfolio with smaller variance whenever $\text{Cov} < 0$ — the algebra of **diversification**.

Correlation: standardised covariance

Definition

For $\text{Var}(X), \text{Var}(Y) > 0$:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} \in [-1, 1].$$

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- $\rho > 0$: positively correlated.
- $\rho < 0$: negatively correlated.
- $\rho = 0$: **linearly** uncorrelated.
- $|\rho| = 1$ iff $Y = aX + b$ deterministically.

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Independence is much stronger than $\rho = 0$.

Two warnings

1. Correlation $\neq$ causation

$\rho(X, Y)$ is symmetric in X, Y . Causal claims need extra structure (experiment, instrument, regression discontinuity) — the central topic of your econometrics class.

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2. Zero correlation $\nRightarrow$ independence

Let $X \sim U[-1, 1]$, $Y = X^2$. By symmetry $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0$, so $\rho = 0$. But Y is a deterministic function of X — maximally dependent.

Correlation captures only the **linear** part of dependence.

Random vectors

Stack $X_1, \dots, X_n$ into a column:

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- Diagonal of $\boldsymbol{\Sigma}$ = variances $\text{Var}(X_i)$.
- Off-diagonal = covariances $\text{Cov}(X_i, X_j)$.
- $\boldsymbol{\Sigma}$ is **symmetric and positive semi-definite**.

Algebra in matrix form

Affine transformations

For constants $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$:

$$\mathbb{E}[A\mathbf{X} + \mathbf{b}] = A\boldsymbol{\mu} + \mathbf{b}, \quad \text{Var}(A\mathbf{X} + \mathbf{b}) = A\Sigma A^\top.$$

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Why we care.

- This is exactly the **algebra of OLS**: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{Y}$.
- Standardised vectors: $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ has mean $\mathbf{0}$ and covariance I .
- In Unit 7 we will define the multivariate normal with this same $(\boldsymbol{\mu}, \Sigma)$.

Example: two-asset portfolio

Returns R_1, R_2 with σ_1, σ_2 and correlation ρ . Portfolio weight $\mathbf{w} = (w_1, w_2)^\top$, $w_1 + w_2 = 1$.

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Variance-minimising weight (one line of calculus):

$$w_1^* = \frac{\sigma_2^2 - \rho \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2}.$$

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For $\rho = -1$ and $\sigma_1 = \sigma_2$, the minimum-variance portfolio is $w_1^* = 1/2$ and **the variance is exactly zero** — perfect hedging.

Conditional expectation function

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- $\mathbb{E}[Y \mid X]$ is a **random variable** (a function of X).
- The map $x \mapsto \mathbb{E}[Y \mid X = x]$ is the **conditional expectation function (CEF)**.

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Why this matters

The CEF is the structural object of regression analysis. “How does the average of Y change with x ?” is the most-asked question in empirical economics.

Example: Akerlof's market for "lemons"

Three equally likely qualities; valuations:

	Seller	Buyer
Lemon	5,000	6,000
Average	6,000	10,000
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- Price $< \$6,000$: must be a lemon. $\mathbb{E}[Y_B \mid Y_S < 6000] = 6,000$.
- Price $\geq \$10,000$: $\mathbb{E}[Y_B \mid Y_S \leq 10000] = 9,000 < 10,000$.

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- Price $\geq \$10,000$: $\mathbb{E}[Y_B \mid Y_S \leq 10000] = 9,000 < 10,000$.

Result. The market for melons **unravels** — **adverse selection**, the classic information-asymmetry result.

Law of iterated expectations (LIE)

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Reading. Average Y over each value of X first, then average those averages over X — you recover $\mathbb{E}[Y]$.

Proof in one line. LOTUS + Fubini:

$$\mathbb{E}[\mathbb{E}[Y | X]] = \int \int y \frac{f_{XY}(x, y)}{f_X(x)} f_X(x) dy dx = \int y f_Y(y) dy = \mathbb{E}[Y].$$

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Generalisation: take out what's known

For any function g of X alone, $\mathbb{E}[g(X)Y] = \mathbb{E}[g(X)\mathbb{E}[Y | X]]$.

Example: compound binomial

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By law of total variance (next slide):

$$\text{Var}(S) = \mathbb{E}[\text{Var}(S \mid X)] + \text{Var}(\mathbb{E}[S \mid X]).$$

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- First term: variation of Y **within** groups defined by X .
- Second term: variation of conditional means **across** those groups.

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Bridge to econometrics

This decomposition is the population analogue of the ANOVA / R^2 decomposition in regression. The “explained” share is what regression captures.

Population OLS as a linear projection

Goal. Approximate Y by an affine function of X : $\hat{Y} = \alpha + \beta X$. Minimise

$$L(\alpha, \beta) = \mathbb{E}[(Y - \alpha - \beta X)^2].$$

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First-order conditions. $\mathbb{E}[Y - \alpha - \beta X] = 0$ and $\mathbb{E}[X(Y - \alpha - \beta X)] = 0$.

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By construction: $\mathbb{E}[U] = 0$, $\text{Cov}(X, U) = 0$, where $U = Y - \alpha - \beta X$.

Why this is the heart of regression

- The OLS coefficient is a ratio of **the same two objects** we computed throughout this unit: a covariance and a variance.
- Empirical OLS replaces Cov, Var by sample analogues.
- The matrix-form generalisation is exactly the $A\Sigma A^\top$ algebra of the random-vector section.

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Looking ahead

- Unit 9 derives OLS as one of three estimation principles (alongside MLE and Method of Moments).
- Unit 10 builds inference on $\hat{\beta}$: t -tests and confidence intervals.
- All of it rests on what we did today.

Take-aways

- 1 Covariance and correlation measure **linear** co-movement. Independence implies zero correlation, but not conversely.
- 2 Random vectors collect $X_1, \dots, X_n$ in $\mathbf{X}$ with mean $\boldsymbol{\mu}$ and covariance $\Sigma \succeq 0$. Affine: $A\mathbf{X} + \mathbf{b}$ has mean $A\boldsymbol{\mu} + \mathbf{b}$, covariance $A\Sigma A^\top$.
- 3 The CEF $\mathbb{E}[Y \mid X = x]$ is the structural object of regression; $\mathbb{E}[Y \mid X]$ is itself a random variable.
- 4 Law of iterated expectations and law of total variance connect conditional and unconditional moments.
- 5 Population OLS: $\beta = \text{Cov}(X, Y)/\text{Var}(X)$. Sample OLS just plugs in sample covariances and variances.

Next unit

Special distributions: Bernoulli, Binomial, Poisson, Uniform, Normal, χ^2 , t , F — including the *multivariate* normal.

Thank you

Questions?

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